

CPU, RTL, and measured-board latency

Python integer kernel

MEASURED

123621.090 ms

Exact C++ native CPU kernel

MEASURED

1777.700 ms

Verilator-generated RTL simulation

MEASURED

2456.708 ms

Pure RTL

DERIVED

54.012600 ms

FPGA board core counter

MEASURED

187144.751 ms

FPGA board system counter

MEASURED

187144.751 ms

Source: [benchmarks/accelerator_benefit/figures/01_cpu_vs_rtl_latency_source.csv](#)

Limitation: Exact C++ is the native CPU baseline; Verilator is separately labeled RTL simulation runtime.